

Median Income and Net Migration across German Districts: A Spatial Econometric Analysis

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Research Question

Question: Is median income associated with net migration across German districts after accounting for regional economic conditions and spatial dependence?

The main focus is on **median income**, while the model also controls for previous migration, GDP per worker, unemployment, education, foreign share, business tax, log population density, rurality and federal-state effects.

Motivation

Migration may respond to income differences, but it can also be shaped by broader local labour-market conditions and spatial clustering between neighbouring districts (Sjaastad, 1962; Harris and Todaro, 1970).

Data and Variables

Item	Description
Spatial unit	400 German districts matched to NUTS-3 regions
Years	2014–2019 in the data; outcome years 2015–2019 after lagging
Dependent variable	Net migration balance
Key variable	Lagged log median income
Lagged controls	Lagged migration and lagged GDP per worker
Other controls	Unemployment, education, foreign population share, business tax, log population density, rurality
Fixed effects	Federal-state effects; year effects used in the panel benchmark
Standard errors	Asymptotic standard errors in the SAR model; district-clustered standard errors in the OLS panel benchmark

Rurality is only available for 2019, so it is treated as a time-invariant structural district characteristic. Data source: INKAR/BBSR (BBSR, 2026); spatial boundaries: Eurostat/GISCO (Eurostat / GISCO, 2021).

Multicollinearity Check

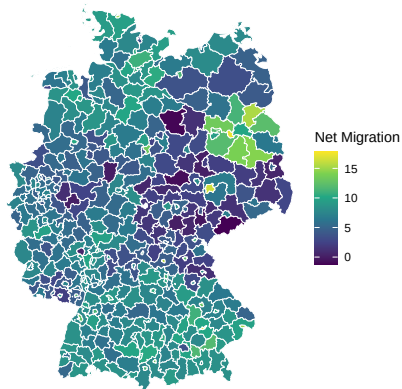
Variable	VIF
Lagged Migration	1.08
Lagged Median Income	5.10
Lagged GDP per Worker	3.43
Unemployment Rate	1.88
Education Level	1.71
Foreign Population Share	3.94
Business Tax	2.21
Log Population Density	7.05
Rurality Index	3.82

Takeaway

The highest VIF values occur for log population density (7.05) and lagged log median income (5.10). All VIF values remain below common concern thresholds, suggesting no severe multicollinearity.

Regional Migration Pattern

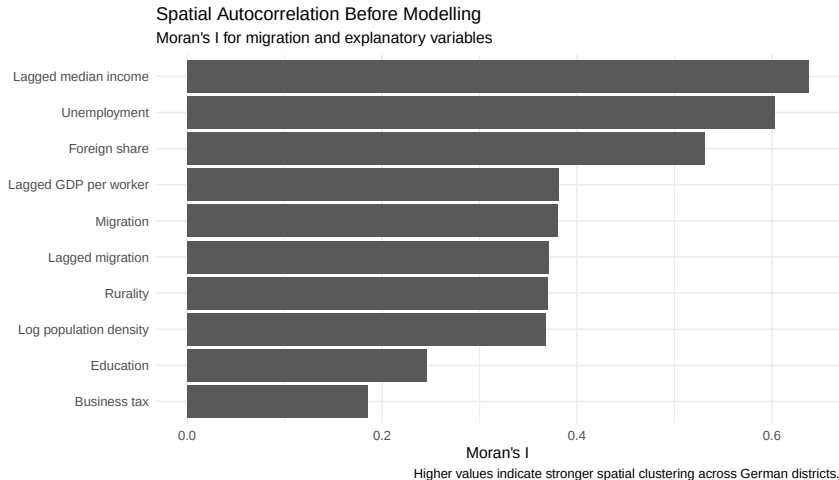
Regional Migration in Germany (2015–2019)
Average net migration balance after using 2014 values for lags



Data Source: INKAR (BBSR)

Average net migration differs clearly across German districts, suggesting that regional location and neighbouring areas may matter. Data: INKAR/BBSR (BBSR, 2026); boundaries: Eurostat/GISCO (Eurostat / GISCO, 2021).

Spatial Autocorrelation: Moran's I



Moran's I measures whether migration and explanatory variables are spatially clustered across neighbouring districts (Cliff and Ord, 1981; Anselin, 1988).

Spatial Model Selection

Model	AIC	Interpretation
SAR	818.4	Best relative fit
OLS	822.5	Non-spatial benchmark Higher AIC Higher AIC
SEM	824.5	
SDM	825.2	

Adjusted Rao score tests from the OLS benchmark

Diagnostic test	p-value	Implication
Adjusted lag test	0.0097	Supports SAR
Adjusted error test	0.447	Does not support SEM

Model choice

SAR is selected because it has the lowest AIC and the adjusted Rao score tests support a spatial lag specification (Anselin, 1988; LeSage and Pace, 2009).

Preferred Spatial Model: SAR

$$Migration_i = \rho WMigration_i + \beta_1 \log(MedianIncome)_i + \beta_2 LaggedMigration_i + \beta_3 \log(GDPworker)_i + X_i\gamma + \delta_s + \varepsilon_i$$

- ▶ W is the row-standardised queen-contiguity spatial weights matrix.
- ▶ $WMigration_i$ captures migration in neighbouring districts.
- ▶ ρ measures spatial dependence in the dependent variable.
- ▶ β_1 is the coefficient of interest for lagged median income.
- ▶ X_i includes unemployment, education, foreign share, business tax, log population density and rurality.

Variables in the SAR model are district-level averages over 2015–2019 after constructing the one-year lags.

Main idea

The SAR model allows migration in one district to depend partly on migration in neighbouring districts.

The SAR specification follows standard spatial autoregressive modelling approaches (Anselin, 1988; LeSage and Pace, 2009; Bivand et al., 2013).

SAR Results

Variable	Estimate	p-value	Result
Spatial lag ρ	0.058	0.012	Significant
Lagged migration	0.903	< 0.001	Significant
Lagged median income	-1.021	0.124	Not significant
Lagged GDP per worker	-0.222	0.622	Not significant
Unemployment	0.012	0.646	Not significant
Education	0.006	0.314	Not significant
Foreign share	-0.029	0.052	Weak / borderline
Business tax	-0.000005	0.979	Not significant
Log population density	-0.030	0.734	Not significant
Rurality	0.0002	0.929	Not significant

Main takeaway

The strongest result is migration persistence. The spatial lag is also significant, while foreign share is only borderline significant and lagged median income is not significant.

Model estimated using the SAR framework described in Anselin (1988) and LeSage and Pace (2009).

SAR Impacts

Because the SAR model includes a spatial lag of the dependent variable, coefficients are interpreted through direct, indirect and total impacts.

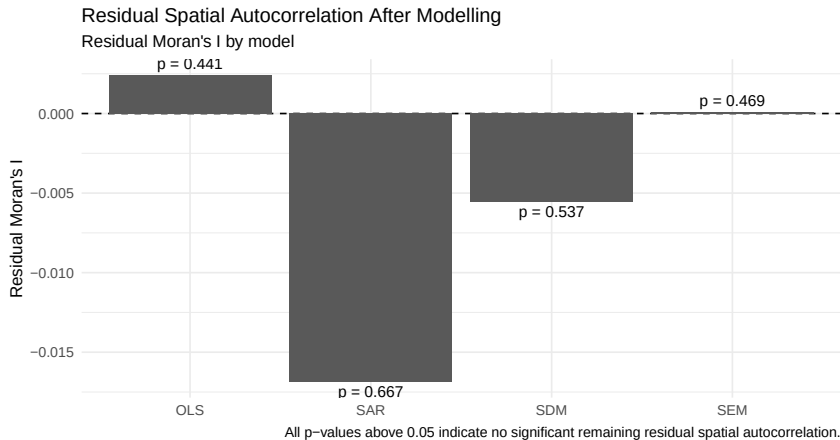
Variable	Direct	Indirect	Total
Lagged migration	0.903	0.055	0.959
Lagged median income	-1.021	-0.063	-1.084
Lagged GDP per worker	-0.222	-0.014	-0.236
Unemployment	0.012	0.001	0.013
Education	0.006	0.000	0.006
Foreign share	-0.029	-0.002	-0.031
Business tax	-0.000005	-0.0000003	-0.000005
Log population density	-0.030	-0.002	-0.032
Rurality	0.0002	0.00001	0.0002

Direct, indirect and total impacts are reported because SAR coefficients do not equal simple marginal effects (LeSage and Pace, 2009).

Interpretation

Indirect effects are small because the estimated spatial lag parameter is small. The main conclusion on median income does not change.

Residual Spatial Autocorrelation



Interpretation

All residual Moran's I p-values are above 0.05, so there is no significant remaining residual spatial autocorrelation (Cliff and Ord, 1981; Anselin, 1988).

Conclusion

- ▶ The map and Moran's I show that migration patterns are spatially clustered across German districts.
- ▶ Among the compared spatial models, the SAR model is preferred based on AIC and adjusted Rao score diagnostics.
- ▶ The SAR model captures significant spatial dependence through neighbouring districts' migration.
- ▶ Lagged median income is not statistically significant in the preferred spatial model.

Main conclusion

There is no robust evidence that higher median income independently explains net migration across German districts after accounting for regional controls and spatial dependence.

Main limitation: the results should be interpreted as associations, not causal effects, because unobserved district-level factors may still influence migration.

References

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Thank you

Questions?

Backup: Main OLS Results

Variable	Estimate	p-value	Result
Lagged migration	0.114	0.075	Weakly significant
Lagged median income	0.445	0.849	Not significant
Lagged GDP per worker	-4.304	0.006	Significant
Unemployment	-0.317	< 0.001	Significant
Education	0.056	0.002	Significant
Foreign share	0.265	< 0.001	Significant
Business tax	-0.0001	0.847	Not significant
Log population density	-0.116	0.716	Not significant
Rurality	-0.0147	0.040	Significant

Backup: SEM Results

Variable	Estimate	p-value	Result
Spatial error λ	0.008	0.919	Not significant
Lagged migration	0.912	< 0.001	Significant
Lagged median income	-1.011	0.131	Not significant
Lagged GDP per worker	-0.101	0.822	Not significant
Unemployment	-0.003	0.902	Not significant
Education	0.005	0.426	Not significant
Foreign share	-0.021	0.161	Not significant
Business tax	-0.000053	0.763	Not significant
Log population density	-0.026	0.771	Not significant
Rurality	0.0008	0.721	Not significant

Backup takeaway

SEM does not improve the model: the spatial error parameter is not significant and the AIC is higher than SAR.

Federal-state dummy coefficients are included but omitted for readability.

Backup: SDM Results

Variable	Estimate	p-value	Result
Spatial lag ρ	0.017	0.812	Not significant
Lagged migration	0.900	< 0.001	Significant
Lagged median income	-1.134	0.088	Weak / 10% level
Lagged GDP per worker	-0.274	0.549	Not significant
Unemployment	-0.012	0.702	Not significant
Education	0.007	0.211	Not significant
Foreign share	-0.008	0.675	Not significant
Business tax	0.000055	0.759	Not significant
Log population density	-0.119	0.200	Not significant
Rurality	-0.001	0.651	Not significant

Backup takeaway

SDM gives a weak negative median-income coefficient, but the spatial lag parameter is not significant and AIC is higher than SAR.

Spatially lagged covariates and federal-state dummy coefficients are included but omitted for readability.